

# SAKTHIVEL VINAYAGAM

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## SUMMARY

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Data Analyst with 3+ years of experience delivering SQL Server and Power BI-driven insights to support operational and quality decision-making. Experienced in analytics engineering, building scalable datasets using complex SQL, and developing dimensional data models for reliable reporting. Known for working closely with business, quality, and IT teams to translate analytical needs into practical data solutions. A proactive self-starter with a strong focus on data accuracy, automation, and exploring AI-driven approaches to improve analysis efficiency. MSc Data Science (Distinction) with a solid foundation in Python analytics and data visualisation.

## TECHNICAL SKILLS & CERTIFICATIONS

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- **Programming & Databases:** Python, SQL (Advanced: CTEs, window functions, complex joins), MS SQL Server, MySQL, R
- **Business Intelligence & Visualisation:** Power BI (DAX, Data Modelling, Power Query), Tableau, Advanced Excel
- **Data Modelling & ETL:** Dimensional data modelling (star schema), data cleaning & validation, ETL pipelines, API integration, automated reporting
- **Machine Learning & Analytics:** Predictive modelling, time series forecasting, NLP
- **Frameworks/Libraries:** Pandas, Numpy, Scikit-learn, TensorFlow, PyTorch, Hugging Face
- **Professional Skills:** Agile/Scrum, cross-functional stakeholder communication, business & data requirements gathering, documentation
- **Certifications:** British Airways – Data Science Simulation (Forage), Tata Group – Data Visualisation Simulation (Forage)

## WORK EXPERIENCE

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**Senior Systems Engineer** | Infosys Ltd | Sep 2021 – Sep 2024 | Chennai, India

- Led the design and delivery of **SQL Server- and Power BI-based analytics solutions** supporting operational and management decision-making across retail and customer operations
- Built **analytics-ready datasets** using advanced SQL and Python to transform large retail datasets (churn, sales, repeat purchases) for reporting and analysis
- Designed **dimensional data models and ETL workflows**, improving data consistency and contributing to ~15% improvement in operational performance
- Automated executive KPI and quality reporting using **Python and Power BI**, reducing reporting effort from ~10 hours to 2 hours per week and improving accuracy by ~30%
- Collaborated closely with business stakeholders and IT teams to **gather analytical requirements** and translate them into scalable dashboards and reporting solutions
- Mentored junior analysts on SQL, Power BI, and analytics best practices, supporting team capability development

**Data Scientist (AI Innovation Project)** | Omdena | Oct 2025 – Dec 2025 | UK, Remote

- Took a leading role in designing and delivering an **AI-driven log normalisation and classification framework**, improving data consistency and usability for large-scale analytics
- Defined a **standardised JSON schema and controlled vocabulary**, reducing data ambiguity and enabling reliable downstream reporting and analysis
- Built **Python-based data validation and transformation pipelines** to ensure data quality, anonymisation, and readiness for analytics and machine learning workflows
- Collaborated with a globally distributed, cross-functional team to deliver an **end-to-end data and AI solution**, contributing to process efficiency and scalable analytics adoption

## SELECTED PROJECTS

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**Mental Health Signal Detection (NLP · Transformers)** | *Python, PyTorch, Hugging Face*

- Fine-tuned DistilBERT on 18K+ Reddit posts to classify mental-health risk levels (Low/Medium/High).
- Improved model performance using cross-validation, achieving macro-F1 improvement of 19% vs baseline logistic regression.

**Weather-Crime Trend Analysis (BI & Data Visualisation)** | *R, ggplot2, Plotly, Leaflet*

- Analysed 16K+ crime + weather records, identifying crime hotspots and seasonal trends.
- Built interactive dashboards enabling correlation exploration for policy insights.

**Stock Market Return Prediction (ML · Finance)** | *Python, Scikit-learn, TensorFlow*

- Engineered technical indicators and evaluated ML models for stock return prediction.
- Selected Linear Regression/SVM based on lowest MSE to support data-driven investment insights.

## EDUCATION

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**MSc Data Science** (Distinction) | University of Essex, Colchester, UK | Oct 2024 – Sept 2025

**Bachelor of Technology, Pharmaceutical Tech** | Anna University (A.C. Tech), Chennai, India | Aug 2017 – Apr 2021

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